



Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Faculty of Engineering Sciences (FES)

Submission Deadline: 18/03/24	LA & Diff. Eqs (MT 102) Spring 2024	Practice Assignment 3
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In Problems 23–28 find an explicit solution of the given initial-value problem.

23. $\frac{dx}{dt} = 4(x^2 + 1), \quad x(\pi/4) = 1$

24. $\frac{dy}{dx} = \frac{y^2 - 1}{x^2 - 1}, \quad y(2) = 2$

25. $x^2 \frac{dy}{dx} = y - xy, \quad y(-1) = -1$

26. $\frac{dy}{dt} + 2y = 1, \quad y(0) = \frac{5}{2}$

27. $\sqrt{1 - y^2} dx - \sqrt{1 - x^2} dy = 0, \quad y(0) = \frac{\sqrt{3}}{2}$

28. $(1 + x^4) dy + x(1 + 4y^2) dx = 0, \quad y(1) = 0$

In Problems 1–24 find the general solution of the given differential equation. Give the largest interval I over which the general solution is defined. Determine whether there are any transient terms in the general solution.

1. $\frac{dy}{dx} = 5y$

2. $\frac{dy}{dx} + 2y = 0$

3. $\frac{dy}{dx} + y = e^{3x}$

4. $3\frac{dy}{dx} + 12y = 4$

5. $y' + 3x^2y = x^2$

6. $y' + 2xy = x^3$

7. $x^2y' + xy = 1$

8. $y' = 2y + x^2 + 5$

9. $x\frac{dy}{dx} - y = x^2\sin x$

10. $x\frac{dy}{dx} + 2y = 3$

11. $x\frac{dy}{dx} + 4y = x^3 - x$

12. $(1+x)\frac{dy}{dx} - xy = x + x^2$

13. $x^2y' + x(x+2)y = e^x$

14. $xy' + (1+x)y = e^{-x}\sin 2x$

15. $y\,dx - 4(x+y^6)\,dy = 0$

In Problems 21–26 solve the given initial-value problem.

21. $(x+y)^2\,dx + (2xy+x^2-1)\,dy = 0, \quad y(1) = 1$

22. $(e^x+y)\,dx + (2+x+ye^y)\,dy = 0, \quad y(0) = 1$

23. $(4y+2t-5)\,dt + (6y+4t-1)\,dy = 0, \quad y(-1) = 2$

24. $\left(\frac{3y^2-t^2}{y^5}\right)\frac{dy}{dt} + \frac{t}{2y^4} = 0, \quad y(1) = 1$

25. $(y^2\cos x - 3x^2y - 2x)\,dx + (2y\sin x - x^3 + \ln y)\,dy = 0, \quad y(0) = e$

26. $\left(\frac{1}{1+y^2} + \cos x - 2xy\right)\frac{dy}{dx} = y(y + \sin x), \quad y(0) = 1$

In Problems 11–14 solve the given initial-value problem.

11. $xy^2 \frac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$

12. $(x^2 + 2y^2) \frac{dx}{dy} = xy, \quad y(-1) = 1$

13. $(x + ye^{y/x}) dx - xe^{y/x} dy = 0, \quad y(1) = 0$

14. $y dx + x(\ln x - \ln y - 1) dy = 0, \quad y(1) = e$

In Problems 15–20 solve the given differential equation by using an appropriate substitution.

15. $x \frac{dy}{dx} + y = \frac{1}{y^2}$

16. $\frac{dy}{dx} - y = e^x y^2$

17. $\frac{dy}{dx} = y(xy^3 - 1)$

18. $x \frac{dy}{dx} - (1 + x)y = xy^2$

21. $x^2 \frac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2}$

22. $y^{1/2} \frac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4$